

Loops

1. While loop

```
In [2]: print('data science')
        print('data science')
        print('data science')
        print('data science')
        print('data science')
```

data science
data science
data science
data science
data science

```
In [3]: i = 1                # initializing
        while i<=5:          # condition
            print('data science')
            i = i + 1         #increment
```

data science
data science
data science
data science
data science

```
In [13]: i = 1               # initializing
          while i<=5:         # condition
              print('data science:',i)
              i = i + 1
```

data science: 1
data science: 2
data science: 3
data science: 4
data science: 5

```
In [15]: i = 5               # initializing
          while i>=1:         # condition
              print('data science')
              i = i - 1       #decrement
```

data science
data science
data science
data science
data science

```
In [16]: i = 5
          while i>=1:
              print('data science:',i)
              i = i - 1
```

```
data science: 5  
data science: 4  
data science: 3  
data science: 2  
data science: 1
```

```
In [17]: i = 100  
         while i>=1:  
             print('data science:',i)  
             i = i - 1
```

```
data science: 100
data science: 99
data science: 98
data science: 97
data science: 96
data science: 95
data science: 94
data science: 93
data science: 92
data science: 91
data science: 90
data science: 89
data science: 88
data science: 87
data science: 86
data science: 85
data science: 84
data science: 83
data science: 82
data science: 81
data science: 80
data science: 79
data science: 78
data science: 77
data science: 76
data science: 75
data science: 74
data science: 73
data science: 72
data science: 71
data science: 70
data science: 69
data science: 68
data science: 67
data science: 66
data science: 65
data science: 64
data science: 63
data science: 62
data science: 61
data science: 60
data science: 59
data science: 58
data science: 57
data science: 56
data science: 55
data science: 54
data science: 53
data science: 52
data science: 51
data science: 50
data science: 49
data science: 48
data science: 47
data science: 46
data science: 45
```

```
data science: 44
data science: 43
data science: 42
data science: 41
data science: 40
data science: 39
data science: 38
data science: 37
data science: 36
data science: 35
data science: 34
data science: 33
data science: 32
data science: 31
data science: 30
data science: 29
data science: 28
data science: 27
data science: 26
data science: 25
data science: 24
data science: 23
data science: 22
data science: 21
data science: 20
data science: 19
data science: 18
data science: 17
data science: 16
data science: 15
data science: 14
data science: 13
data science: 12
data science: 11
data science: 10
data science: 9
data science: 8
data science: 7
data science: 6
data science: 5
data science: 4
data science: 3
data science: 2
data science: 1
```

In []:

can we use multiple while loop || nested while loop

```
In [18]: i = 1

while i<=5:
    print('data science') # when we mention end then new line will not create
    j = 1
    while j<=4:
```

```

    print('technology')
    j = j + 1

    i = i + 1
    print()

```

the output which we got is very lengthy but how to make them one line Lets ref

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

data science
technology
technology
technology
technology

```

```

In [20]: i = 1
        while i<=5:
            print(' datascience', end=" ") # when we mention end then new line will not cre
            j = 1
            while j<=4:
                print(' technology', end=" ")
                j = j + 1

            i = i + 1
            print()

```

```

datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology
datascience technology technology technology technology

```

```
In [21]: # Lets use while loop usig some numbers
i = 1

while i <= 2 :
    j = 0
    while j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2
0 2 4
```

```
In [22]: i = 1
while i <= 4 :
    j = 0
    while j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

2. ForLoop

```
In [26]: name = 'nit'

for i in name:
    print(i)
```

```
n
i
t
```

```
In [27]: name1 = [1,3.5,'hallo'] #i want print the value individually

for i in name1:
    print(i)
```

```
1
3.5
hallo
```

```
In [28]: for i in [2, 3, 7.8, 'hi']:
    print(i)
```

```
2
3
7.8
hi
```

```
In [29]: range(5)
```

```
Out[29]: range(0, 5)
```

```
In [30]: for i in range(5):  
         print(i)
```

```
0  
1  
2  
3  
4
```

```
In [31]: for i in range(2,5):  
         print(i)
```

```
2  
3  
4
```

```
In [32]: for i in range(1,10,3):  
         print(i)
```

```
1  
4  
7
```

```
In [33]: for i in range(1,21):  
         print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20
```

```
In [35]: # print the value which is divisible by 5  
         for i in range(1,51):  
             if i%5==0 :  
                 print(i)
```

5
10
15
20
25
30
35
40
45
50

```
In [36]: # print the value which is divisible by 5 i dont want that value  
for i in range(1,51):  
    if i%5!=0 :  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
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32
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44
46
47
48
49

In []:

LETS DISCUSS ABOUT 3 KEYWORDS -- BREAK || CONTINUE || PASS

```
In [42]: for i in range(1,11):  
         print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

```
In [45]: for i in range(1,11):  
         if i == 6:  
             break  
         print(i)
```

```
1  
2  
3  
4  
5
```

```
In [46]: for i in range(1,11):  
         if i == 3:  
             continue  
         print(i)
```

```
1  
2  
4  
5  
6  
7  
8  
9  
10
```

```
In [47]: for i in range(1,11):  
         if i == 6:  
             continue  
         print('hello : ',i)
```

```
hello : 1
hello : 2
hello : 3
hello : 4
hello : 5
hello : 7
hello : 8
hello : 9
hello : 10
```

```
In [48]: for i in range(1,11):
```

```
Cell In[48], line 1
    for i in range(1,11):
                        ^
SyntaxError: incomplete input
```

```
In [49]: for i in range(1,11):
        pass
```

you need to print the number from 1 to 50 but dont print the number which is divisible by 3 or 5

```
In [50]: for i in range(1,51):
```

```
    if i%3 == 0:
        print(i)
print('end')
```

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end
```

```
In [53]: for i in range(1,51):
```

```
    if i%3 != 0:
        print(i)
print('end')
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

```
In [54]: for i in range(1,51):
          if i%3 == 0 or i%5 == 0:
              continue
          print(i)
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49

```
In [55]: for i in range(1,50):  
         if i%3 == 0 or i%5 == 0:  
             continue  
         print(i)  
         print('end')
```

1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end

```
In [58]: for i in range(1,51):  
        if (i%2 == 0):  
            #print('even')  
            continue  
        else:  
            print(i)  
print('bye')
```

```

1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
bye

```

In []:

PRINTING PATTERN IN PYTHON

```

In [59]: print('# # # #')
          print('# # # #')
          print('# # # #')
          print('# # # #')

```

```

# # # #
# # # #
# # # #
# # # #

```

```

In [60]: for i in range(1,5):
          i=i+1
          print('# # # # ')

```

```

# # # #
# # # #
# # # #
# # # #

```

```

In [61]: for i in range(1,5):
          if i<=5:
              print('# # # #')

```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [62]: for j in range(4):  
         print('#')
```

```
#  
#  
#  
#
```

```
In [63]: for j in range(4):  
         print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```